

Week 7 Assignment — Build a Production-Style RAG System from Scratch

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Abstract

In this project, we built a Retrieval-Augmented Generation (RAG) system to answer questions from academic policy documents. The main goal was to reduce wrong answers (hallucination) from normal language models. The system retrieves relevant parts of documents and uses them to generate correct answers. When we compared normal LLM answers with RAG answers, we observed that RAG gives more accurate and reliable results. This shows that RAG is useful for real-world applications where correct information is important (Lewis et al., 2020).

Problem Statement

The SLU academic policy document contains important information like GPA requirements, attendance rules, and academic probation. But these documents are long and difficult to search manually.

If we use only a language model, it gives general answers and sometimes wrong information. For example, it may say GPA is 2.0 instead of the correct value. This happens because LLMs rely on learned knowledge and may hallucinate (Lewis et al., 2020).

So, the problem is:

How to build a system that gives correct answers based on real documents instead of guessing?

To solve this, we used a RAG system which combines retrieval and generation.

System Design

The system works in the following steps:

1. Load the PDF documents
2. Clean the text
3. Split the text into chunks
4. Convert chunks into embeddings

5. Retrieve the most relevant chunks using cosine similarity
6. Pass retrieved context to the LLM
7. Generate final answer

The system retrieves top 2 chunks, which helps improve answer quality by providing better context (Lewis et al., 2020).

Dataset Description

The dataset contains academic policy documents.

- Total pages: 13
- Average length: ~1706 characters

The text was cleaned by removing unnecessary words like “Question” and “Answer”.

Chunking Strategies

Two types of chunking were used:

1. Sentence Chunking

- Breaks text into small parts
- More precise
- But sometimes loses full meaning

2. Paragraph Chunking

- Keeps full context
- Better understanding
- Slightly larger chunks

Result:

Paragraph chunking worked better because it keeps more context and gives complete answers. This supports the idea that preserving semantic meaning improves retrieval performance (Reimers & Gurevych, 2019).

Question: What GPA is required?

■ Paragraph Chunking

Students in good standing have a grade point average (GPA) of 2.5 or higher.

■ Sentence Chunking

The minimum cumulative GPA requirement for the receiving program is 2.5.

Question: What happens if attendance falls below 75%?

■ Paragraph Chunking

Students are placed on program probation if they fail to meet the requirements for academic good standing. They may be required to repeat all co-requisite courses or the full level for Academic English students.

■ Sentence Chunking

You will be on academic probation for the next semester, even if you progress to the next level of study.

Model Choices

Embedding Model

- all-mpnet-base-v2
- Used to convert text into vectors
- Helps in finding similar meaning

Embedding models help in semantic search by comparing meaning instead of exact words (Reimers & Gurevych, 2019).

LLM

- TinyLlama-1.1B
- Used to generate answers

Lightweight language models like LLaMA-based models are efficient and suitable for such tasks (Touvron et al., 2023).

Results Comparison (RAG vs LLM-Only)

Query 1: GPA Requirement

- LLM Answer:
GPA = 2.0 (Incorrect)
- RAG Answer:
GPA = 2.5 (Correct)

Question: What GPA is required to remain in good academic standing?

✖ LLM Answer

To remain in good academic standing, a student must maintain a GPA of 2.0 or higher. Example: If a student has a GPA of 2.5, they will be placed on academic probation. If a student has a GPA of 2.0, they will be placed on academic warning. If a student has a GPA of

✔ RAG Answer

Students in good standing have a GPA of 2.5 or higher.

Retrieved Context

Context 1:

of study. Students in good standing have a grade point average (GPA) of 2.5 or higher. You should be familiar with your program's progression requirements. It is a good idea to know your current GPA and speak with your instructors, your academic advisor, or the Associate Director for Pathway Programs if you have any questions about your progress in your classes. Grade Appeals Students may appeal course grades by completing the grade appeal process. Meet with your academic advisor or a member of the academic leadership team to complete the grade appeal form that begins the appeal process. Time in Academic English and Pathway Programs A student beginning in Academic English Level 1 and making normal progress through a 2-

Context 2:

Standing As a student in an INTO SLU program, you are expected to make good progress towards your major area of study by remaining in good standing. See Appendix A for a list of progression requirements by program, which identify the specific requirements to be in good standing.

Query 2: Attendance Below 75%

- LLM Answer:

Gives general explanation (Incorrect)
- RAG Answer:

Student will be on academic probation next semester (Correct)

Question: What happens if attendance falls below 75%?

✖ LLM Answer

If attendance falls below 75% for a single class, the school will be required to provide a written report to the district office. The report will include a detailed explanation of the reasons for the low attendance, the steps taken to address the issue, and the results of any interventions. The report will also include recommendations for improving attendance in the future.

✔ RAG Answer

Students are placed on program probation if they fail to meet the requirements for academic good standing. They may be required to repeat all co-requisite courses or the full level for Academic English students.

Retrieved Context

Context 1:

Students are required to attend 75% or more of all class meetings during the semester to be in good standing at INTO Saint Louis University. This means you can miss no more than 25% of class sessions. If you are more than 30 minutes late to class or if you leave class for more than 30 minutes, you are counted absent for the full period. Check with each of your instructors for information specific to your classes. If you miss more than 25% of class sessions for any reason you will be on academic probation next semester, even if you progress to the next level of study. You should attend all class sessions to get the most educational benefit from

Context 2:

Probation If you fail to meet the requirements for academic good standing, you will be placed on program probation. Probation is a warning that indicates you are not currently meeting program progression requirements. It may also mean that you need additional support to be successful. Reason for program probation include: • Your GPA falls below your program's requirement, • You earn grades below a C (B for major courses), • You exceed the number of absences allowed by the program, or • You do not pass your Pathway portfolio. During a probation semester in which you are repeating courses, you will be required to repeat all co requisite courses (or the full level for Academic English students). You may repeat

Query 3: Academic Probation

- LLM Answer:

Generic definition (Not accurate)
- RAG Answer:

Warning that student is not meeting program requirements (Correct)

Question: What is academic probation?

✗ LLM Answer

Academic probation is a condition that a student must meet in order to avoid being expelled from the university. It is a warning that the student's academic performance is not up to par and that they need to improve their grades and attendance. If the student fails to meet the conditions of academic probation, they may be placed on academic probation for a specific period of time

✓ RAG Answer

Academic probation is a warning that indicates you are not currently meeting program progression requirements. It may also mean that you need additional support to be successful.

Retrieved Context

Context 1:

Probation If you fail to meet the requirements for academic good standing, you will be placed on program probation. Probation is a warning that indicates you are not currently meeting program progression requirements. It may also mean that you need additional support to be successful. Reason for program probation include: • Your GPA falls below your program's requirement, • You earn grades below a C (B for major courses), • You exceed the number of absences allowed by the program, or • You do not pass your Pathway portfolio. During a probation semester in which you are repeating courses, you will be required to repeat all co requisite courses (or the full level for Academic English students). You may repeat

Context 2:

a level up to two times before dismissal. If you are granted reinstatement after being dismissed, you may repeat the level a third and final time. Dismissal If you are on probation for two semesters and do not meet the terms of your probation at the end of your third semester, you may be dismissed. Dismissal Appeals You may appeal the decision the first time you are dismissed from our program. You will need to submit the form emailed to you by the Academic Director and write a letter in support of your case. You may only appeal the first dismissal; if you do not return to good standing in the semester after dismissal, you will be dismissed and will

These results show that RAG improves accuracy by using actual document context instead of general knowledge (Lewis et al., 2020).

GPA Question

- Paragraph: GPA = 2.5
- Sentence: GPA = 2.5

Attendance Question

- Paragraph: Academic probation
- Sentence: Academic probation

Both methods worked, but paragraph chunking provided better context and clearer answers.

Observations

- RAG answers are more accurate than LLM-only
- Retrieval quality is very important
- Chunking affects performance
- LLM alone gives generic answers
- RAG uses real data and gives correct answers

This shows that retrieval plays a major role in improving LLM performance (Lewis et al., 2020).

Business Impact

The RAG system provides several advantages:

- Accuracy: Responses are grounded in real documents
- Efficiency: Faster retrieval of policy information
- Reliability: Reduced hallucination improves trust
- Scalability: Applicable across domains such as HR, legal, and support systems

These benefits make RAG systems valuable for enterprise knowledge management and decision-making applications. RAG systems are useful in real-world applications where accuracy is important (Lewis et al., 2020).

Conclusion

This project shows that RAG systems are better than normal LLMs. By using real document data, the system gives more accurate and reliable answers. This makes RAG useful for applications where correctness is critical (Lewis et al., 2020).

References

- Bathula, C. P. (2026). *AA-5750: Contemporary Issues in Analytics*. GitHub
https://github.com/ChandraPrakash-Bathula/AA-5750-Contemporary_Issues_In_Analytics
- Lewis, P., Perez, E., Piktus, A., et al. (2020). *Retrieval-augmented generation for knowledge-intensive NLP tasks*. *Advances in Neural Information Processing Systems*.
<https://arxiv.org/abs/2005.11401>
- Reimers, N., & Gurevych, I. (2019). *Sentence-BERT: Sentence embeddings using Siamese BERT-networks*. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. <https://arxiv.org/abs/1908.10084>
- Touvron, H., et al. (2023). *LLaMA: Open and efficient foundation language models*.
<https://arxiv.org/abs/2302.13971>